🔑 **Answer Key — Teacher Copy · Lesson 4-4**

MSTAR PRACTICE WORKSHEET

Lesson 4-4: Find and Compare with Percentages

Grade 6 Mathematics · Standard 6.AT.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

What is 20% of 150?

A. 20

B. 30

C. 35

D. 75

20% = 0.20. Multiply: 0.20 × 150 = 30.

Question 2 SELECTED RESPONSE — CORRECT: B

What is 50% of 84?

A. 34

B. 42

C. 50

D. 168

50% = 0.50. Multiply: 0.50 × 84 = 42. (Or simply divide by 2: 84 ÷ 2 = 42.)

Question 3 SELECTED RESPONSE — CORRECT: B

What is 10% of 320?

A 3.2

B 32

C 64

D 320

10% = 0.10. Multiply: 0.10 × 320 = 32. (Or move the decimal one place left.)

Question 4 SELECTED RESPONSE — CORRECT: B

What percent of 50 is 15?

A 15%

B 30%

C 35%

D 50%

15/50 → 50 × 2 = 100, so 15 × 2 = 30. That makes 15/50 = 30/100 = 30%.

Question 5 SELECTED RESPONSE — CORRECT: C

An arcade prize costs 240 tickets. If you have a coupon for 15% off, how many tickets do you save?

A 15 tickets

B 24 tickets

C 36 tickets

D 204 tickets

15% of 240 = 0.15 × 240 = 36 tickets saved.

Question 6 SELECTED RESPONSE — CORRECT: C

A student scored 72 out of 90 on a test. What percent did they score?

A. 72%

B. 75%

C. 80%

D. 90%

72/90 simplifies to 8/10, and 10 × 10 = 100, so 8 × 10 = 80. That makes 72/90 = 80/100 = 80%.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

At West Middle School, 45% of the 360 sixth-grade students participate in after-school sports. How many sixth-grade students participate in sports?

A. 144 students

B. 160 students

C. 162 students

D. 180 students

45% of 360 = 0.45 × 360 = 162 students.

- **A:** That is 40% of 360. Multiply by the actual rate: 0.45 x 360.
- **B:** That is an estimate rather than the exact product. 0.45 x 360 comes to 162.
- **D:** That is 50% of 360. The rate is 45%, a little less than half.

Part B — correct answer: A

Which calculation method accurately justifies this solution in Part A?

A. Multiply the decimal representation of the percent by the whole: 0.45 × 360 = 162.

B. Subtract 45 from 360: 360 - 45 = 315.

C. Divide 360 by 45: 360 ÷ 45 = 8.

D. Multiply 360 by 4.5: 360 × 4.5 = 1,620.

Finding a percent of a number requires multiplying the rate per hundred (0.45) by the total population (360).

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Compare Percent Discounts

> Store A offers a $120 skateboard at 30% off. Store B offers the same $120 skateboard with a $40 instant manufacturer coupon.

Which store offers the lower final price? Justify your decision by showing the savings and final price at both stores.

Model answer: Store B offers the lower final price ($80 vs $84). Store A discount is 30% of $120 = $36 (final price $84). Store B discount is $40 (final price $80). Store B saves $4 more.

2 points	Student calculates Store A discount as $36 (0.30 × 120) with final price $84, calculates Store B final price as $80 ($120 - $40), and concludes Store B is cheaper by $4.
1 point	Student correctly calculates discounts for both stores but makes an arithmetic error on final price or fails to state which store is cheaper.
0 points	Student chooses a store without supporting mathematical calculations.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A class has 32 students. 25% are in the math club, 50% are in the reading club, and 12.5% are in both. How many students are in each group? Can the total of all three groups be more than 32? Explain why or why not.

Model answer: Taking each percent of the same 32 students gives 25% = 32 ÷ 4 = 8 in math club, 50% = 32 ÷ 2 = 16 in reading club, and 12.5% = 32 ÷ 8 = 4 in both. Adding 8 + 16 + 4 = 28 is misleading, because the 4 students in both are already inside the 8 and the 16 — counting them a third time overlaps, and the number of different students in at least one club is 8 + 16 − 4 = 20.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.
